University of Hertfordshire UH

Department of Engineering

CFD Analysis of an F1 front wing

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1. Introduction

Formula one is the most advanced racing in all of motorsport. Every competitor in this sport focuses on increasing the speed and stability of their car both in straights as well as the corner sections of the circuits around the world. Drag and downforce are vital for the performance of the car. Drag and efficiency are inversely proportional each other. Additionally, Downforce increases the stability of the car at high speeds. Simulation helps in designing the parts of the car such that the wing produces enough downforce that the car is stable but not slow and the drag is minimised with certain features on the body which in turn increases the performance and the handling on the road. Furthermore, this technology of decreasing drag and increasing downforce to a certain amount can be used in road vehicles as the companies compete to produce the most fuel efficient vehicles and also with best-in –class power.

The aim of this assignment is to conduct an CFD analysis of an formula one front wing using Star-CCM+ (an industrial software used by major OEM's) and create an aero-CFD computational model of the wing to decide the best mesh size to be used and flow visualization for the wing at 200mph and 259.5mph (individual speed).

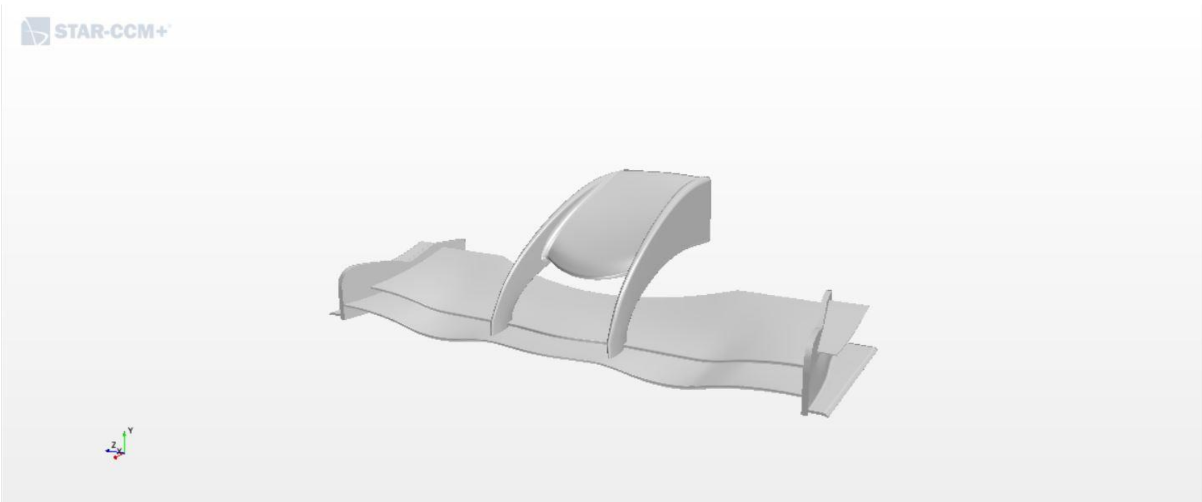

Fig 1.1: 3d Model of the front wing

1.1 Approach towards solution

- The model wing is imported to Star-CCM+ in which the surface is repaired and ready to mesh
- Defining a region or block in which the model is placed. This acts as a wind tunnel with the **prescribed velocity of air of 259.5mph**

- Defining the fluid (air) and its properties to be used in the simulation and the initial conditions such as velocity and pressure.
- Meshing of the model is carried out by using different base sizes at first for the same speed i.e., 200mph to select an appropriate choice for individual speed by noticing the time taken to create the volumetric mesh and the variation in results.
- Run the simulation and tabulate the data in terms of Drag (N), Downforce (N) and Average Pressure (Pa) for different base sizes.
- Choosing the appropriate base size for our individual speed looking upon the residuals and the results plots.
- Running the simulation for the individual speed (259.5mph) and tabulating the results for analysis.

2. Modelling and description

Creating an aero-CFD computational model has its procedure of things that needs to be sorted before running it. This plays a vital role in the type of results that are needed to be acquired to analyse and design a better wing to increase downforce and decrease drag.

2.1 Defining Region

The aero model of the wing can be produced by using only half of the wing as both the halves are symmetrical and we can define a region for one half of the wing. Then the results can be matched to the other half using the symmetry plane.

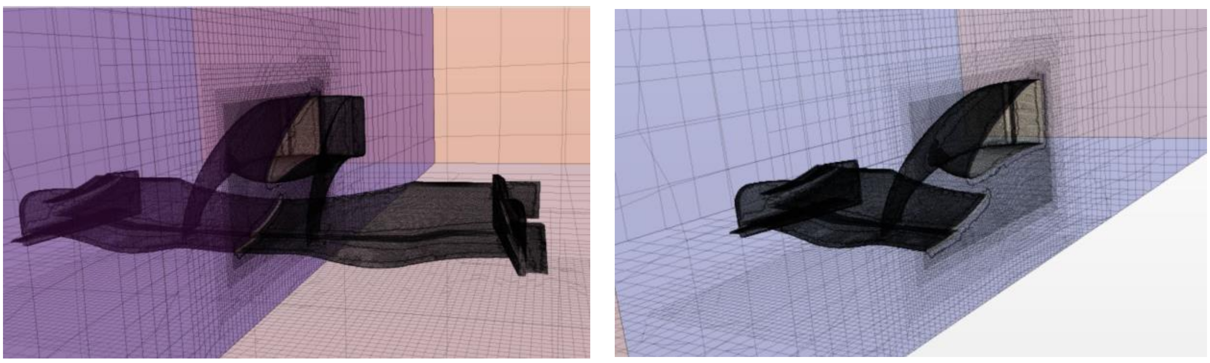

Fig 2.1: Creating a half wing region using symmetry plane

Additionally, the user needs to define another region which acts as velocity tunnel around the wing. The size of this region is bigger than the wing to simulate real world environment. Certain planes are distinguished in this region to carry out the simulation.

2.1.1 Established Regions

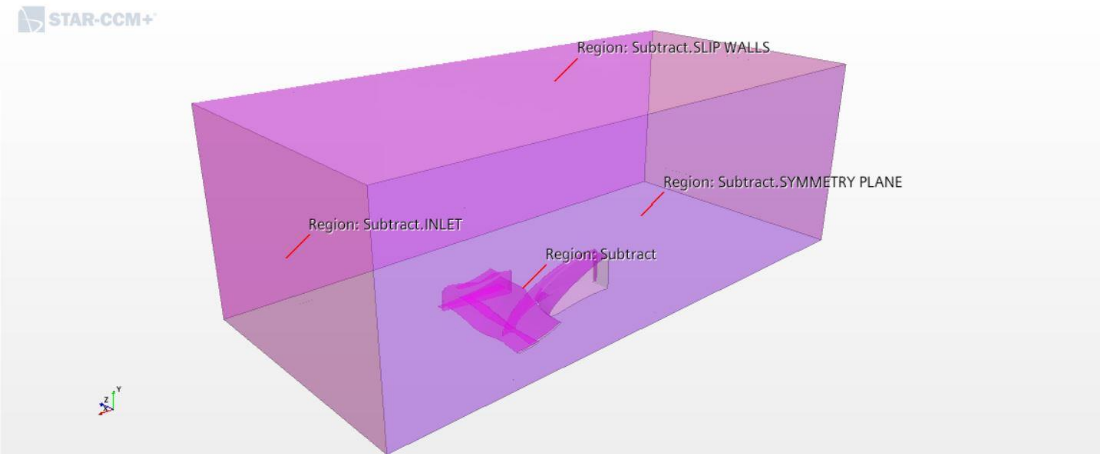

Fig 2.2: Regions defined using a block

Inlet: The plane which is in front of the wing and lets the software know the entry of air flow. Defining this region, the initial value of the velocity is provided which means the fluid flows through the inlet plane at the given velocity.

Outlet: This provides the outlet for the fluid from the region. It is defined exactly opposite to the inlet plane.

Symmetry plane: Considering only half of the wing is done by the symmetry plane. This divides the wing into two equal halves and provides results for one half and helps in reproducing the results to the other half

Slip walls: The friction less planes which does not affect the flow of fluid while travelling from inlet to the outlet. It provides a boundary in which the fluid can travel.

Floor: This plane emulates the ground in the region. The base of both the model and the block is the floor plane.

2.2 Flow and fluid specifications

- Steady flow is considered as the velocity remains the same at any point of time and it emulates the conditions inside a wind tunnel.
- As in the wind tunnel, there are no density variations of the pressure of the fluid, hence, segregated flow and constant density for the fluid
- The density of the working fluid remains the same during its motion which makes it incompressible. In our case the working fluid is gas to replicate the real world conditions.
- Kappa-Omega turbulence model is being used in our simulation as it is better in simulating flow around the mesh to with the parameters we have chosen.

2.3 Mesh Independency study

Meshing of the wing is a vital step in the simulation as it can affect the results. Hence, we studied the results of different mesh sizes with the same velocity of 200mph. This study helps us to prove that the results of the simulation is dependent on the mesh size selected. Therefore, a relevant mesh size is chosen by analysing the results of various sizes to run the simulation for the independent speed of 259.5mph. The base sizes considered for this study are 0.4m, 0.5m, 0.6m, 0.7m, 0.8m, 0.9m, 1m. The following plots shows us the variation of results (Drag, Downforce and overall pressure) with respect to the before said mesh sizes.

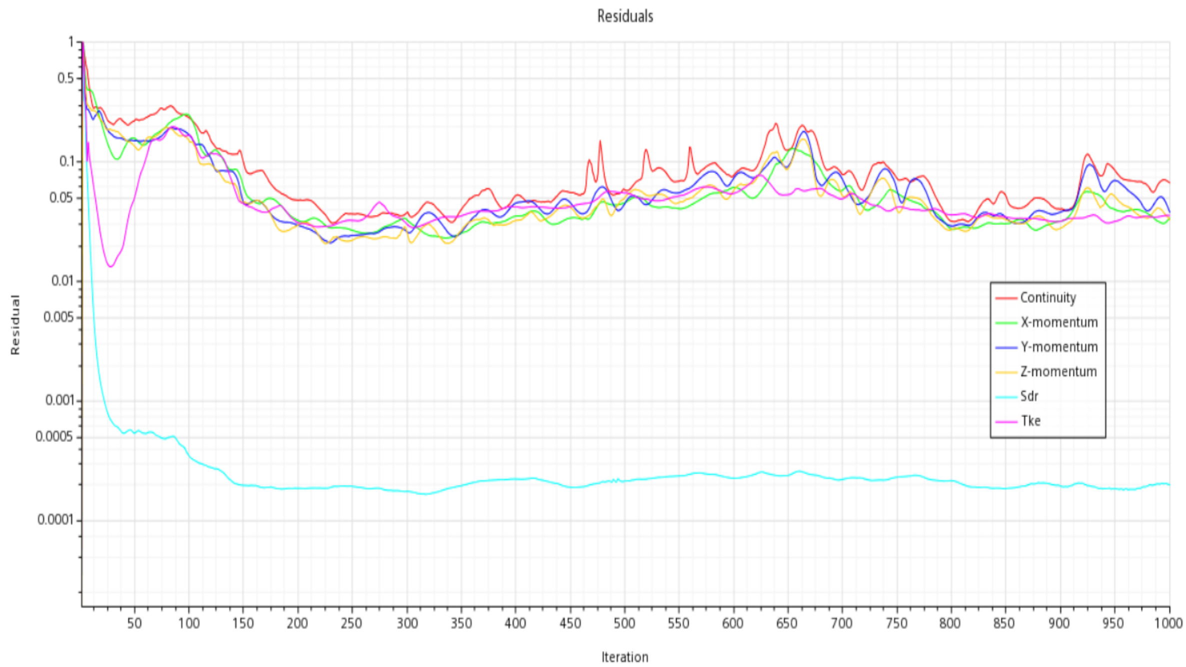

Fig 2.3: Residual graph of the simulation run at 0.5m base size

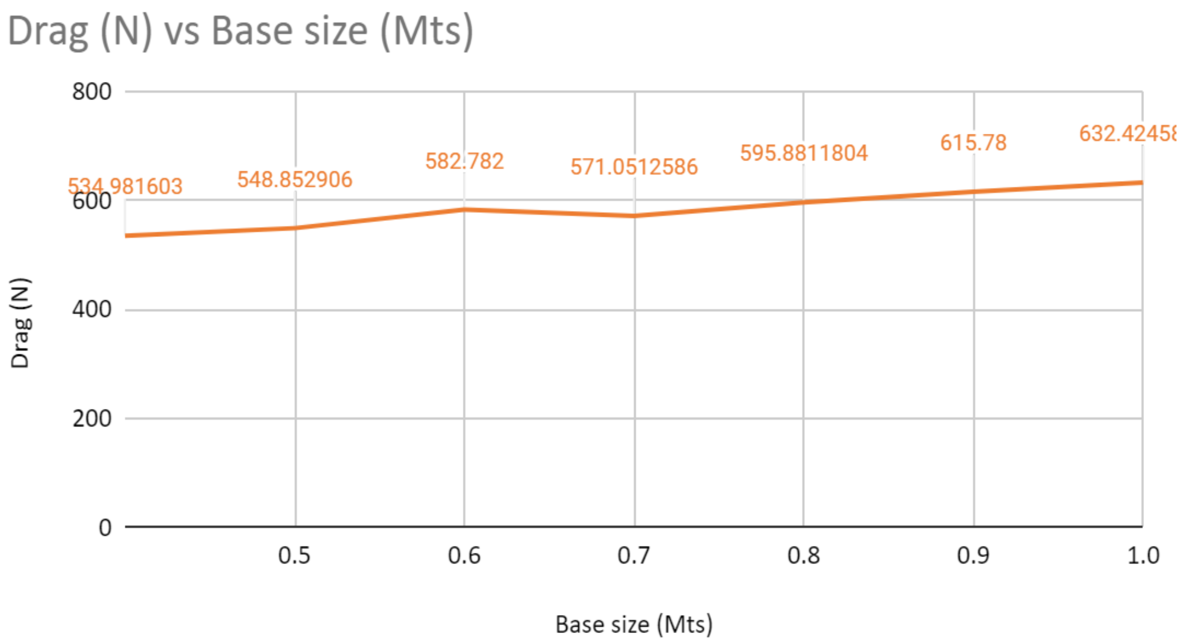

Fig 2.4: Graph of Drag (N) vs different mesh sizes

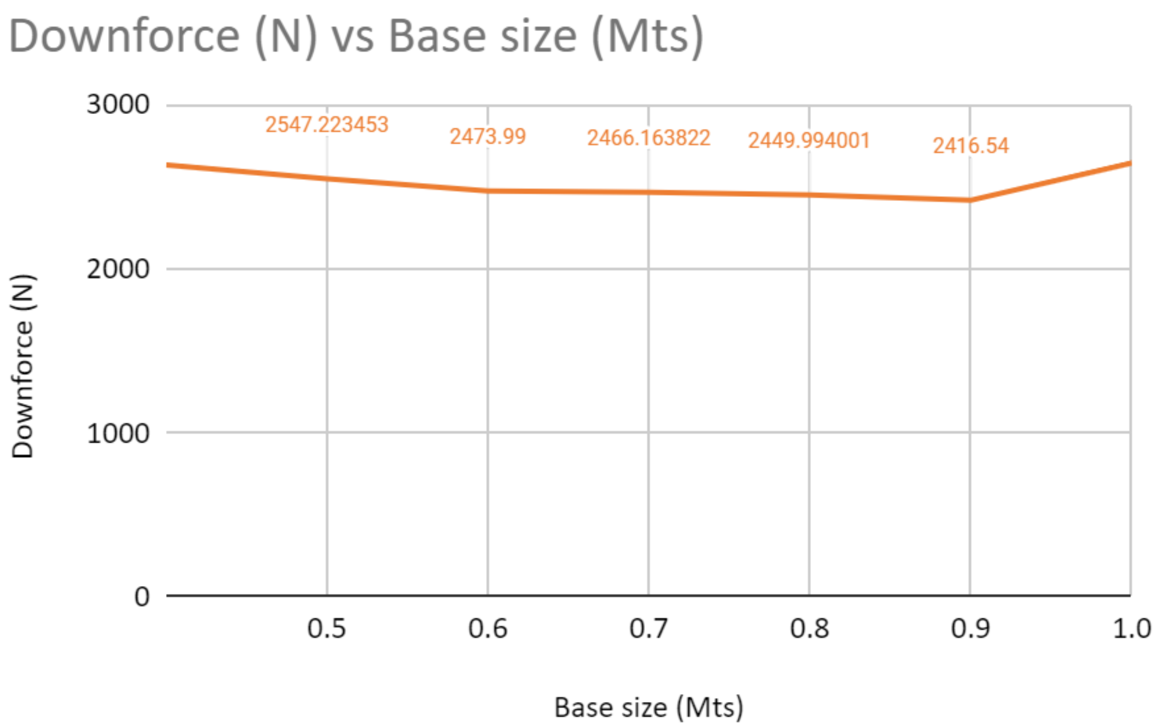

Fig 2.5: Graph of Downforce (N) vs different mesh sizes

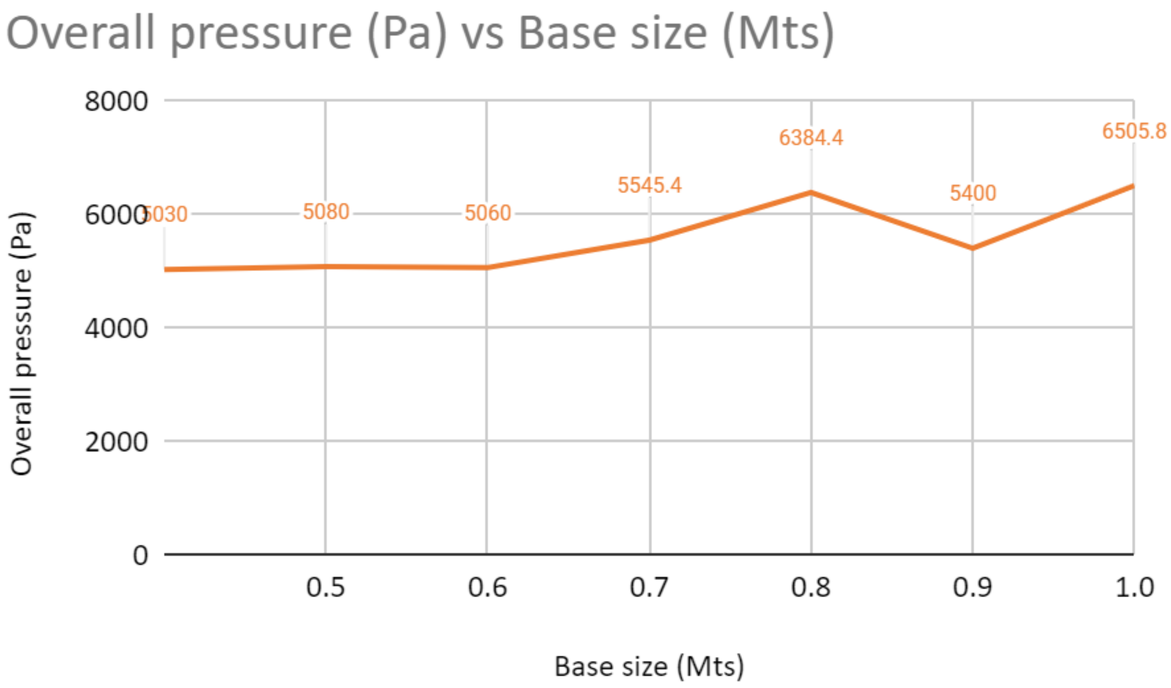

Fig 2.6: Graph of Overall Pressure (Pa) vs different mesh sizes

3. Results and Analysis

Conducting simulation of the formula one wing at individual speed of 259.5mph with the selected mesh size 0.5m gives out the following results.

3.1 Residuals

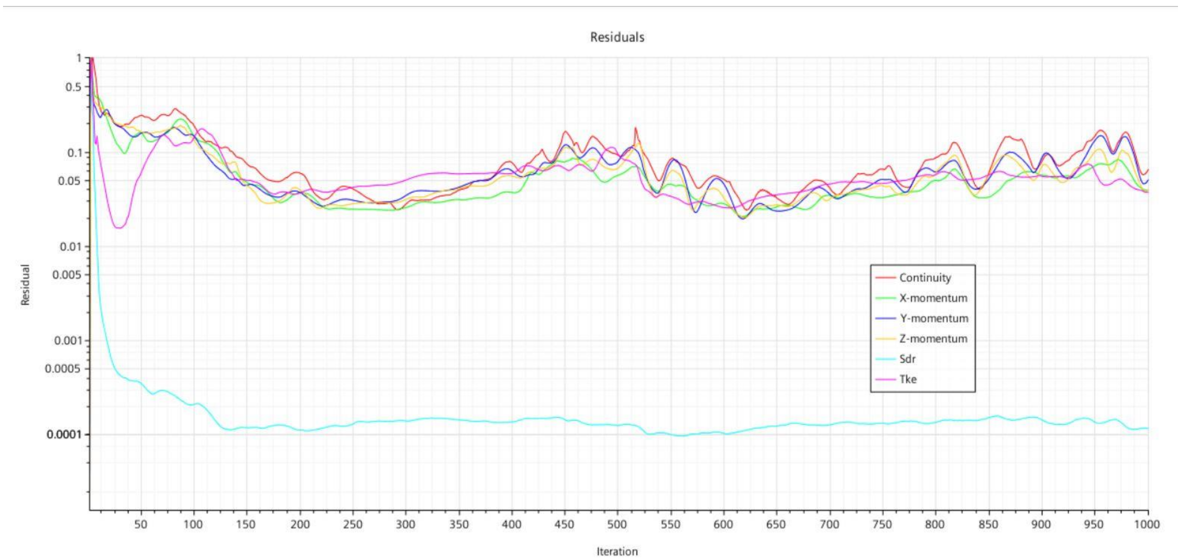

Fig 3.1: Residual plot for the simulation

3.2 Drag (N) and Downforce (N)

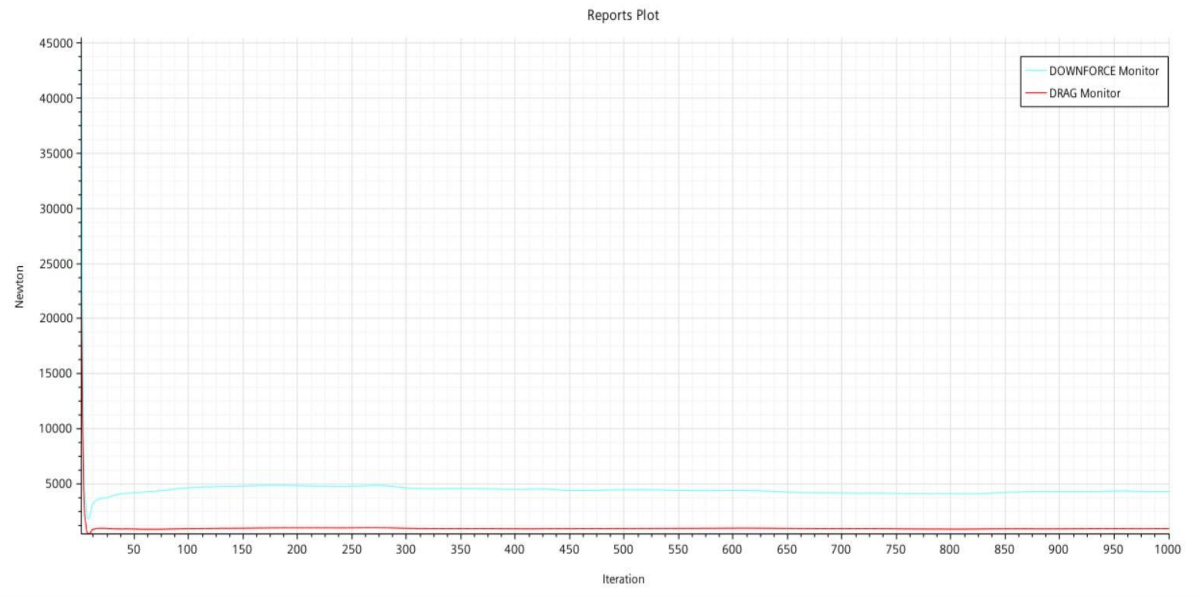

Fig 3.2: Graph showing Drag (N) and Downforce (N) vs Iterations

Drag and Downforce given in Newton, tabulates the different values at different iterations. We see a dip in downforce at the start of the simulation. At around the 100th iteration the downforce line becomes linear to the fluid flow. Simultaneously, the drag monitor also follows the same pattern of decreasing at the start but becomes linear before the downforce does. This irregularity of increasing and following a linear pattern is due to the flow of air when it first hits the wing and then it flows without having to change its density.

3.3 Streamline representation of Pressure (Pa) and Velocity (m/s)

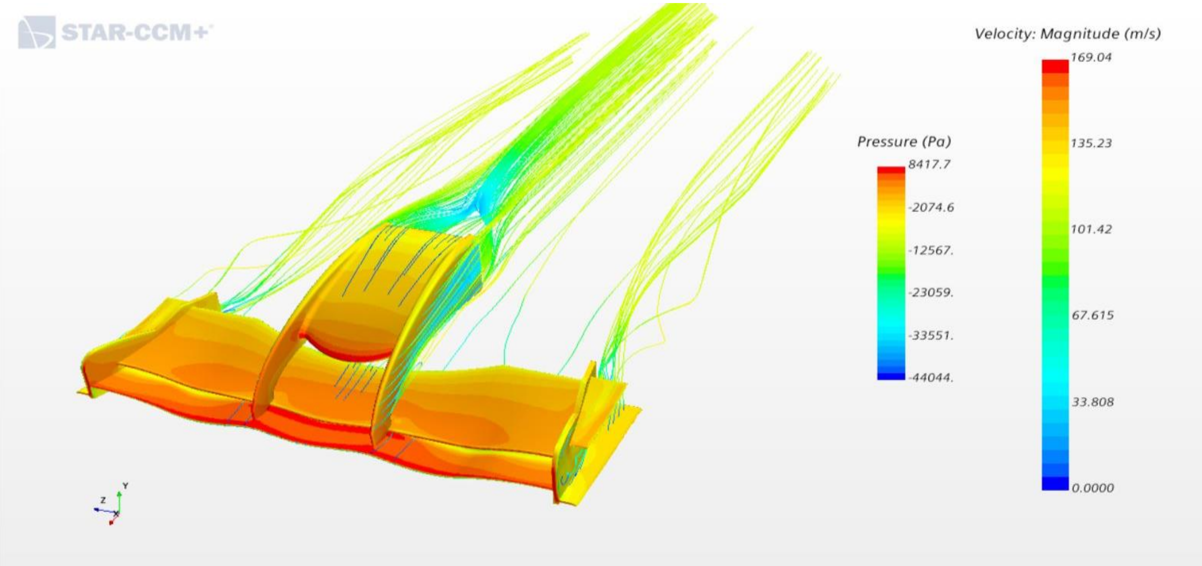

Fig 3.3: Streamline representation of Velocity (m/s) and Pressure (Pa)

Streamline representation creates a whole new dynamic for understanding the fluid flow in the wing. As discussed before, the drag and downforce decreases in the beginning of the simulation. This can be seen even with velocity and pressure in the abouve figure. The pressure on the edges of wings is comparatively less than the pressure on the leaves of the wing. The fluid is distributed by the edges which reduces the pressure. On the other hand, Velocity decreases as it hits the wing and flows towards the exit. The streamlines shows the variation in velocity. The change from red to green and blue shows the decrease in its value. We can also observe the teal region behind the nose where the streamlines converge and creates air vortices. This can be avoided by directing the air to other parts of the car

3.4 Vector representation of Velocity (m/s)

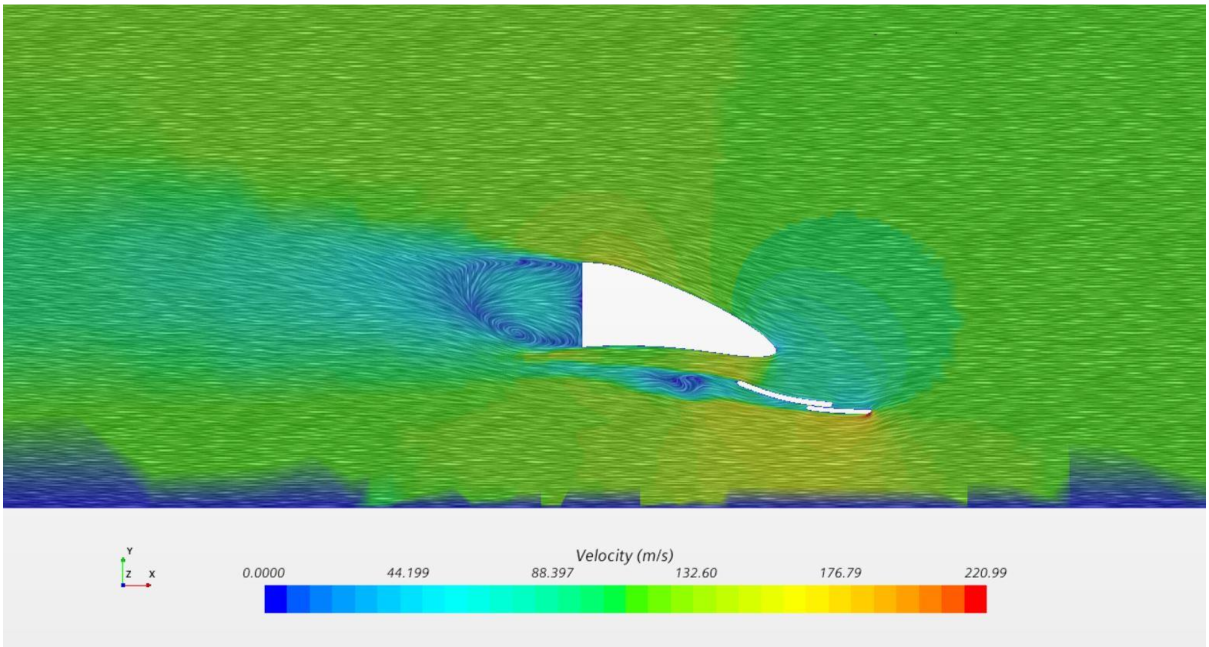

Fig 3.4: Vector representation of Velocity (m/s)

The teal region behind the nose represents the air vortices created by the convergence of the streamlines. The maximum velocity region is under the wing illustrated by the colour red. As the wing distributes air coming from the inlet, the region under the nose experiences high pressure as it should withstand the pressure on top of the wing as well the pressure from the air coming towards it. Adding more components in the wing such as canards, lips can reduce this high stress point and also prevent the turbulence behind the nose.

4. Conclusion

The analysis of the above results depicts the values of the drag and downforce experienced by the front wing at 259.5mph in a wind tunnel We can observe the advantage of using the base size of 0.5m in the simulation results. The velocity magnitude of 200mph was used in the mesh independency study so that the relevant mesh size was selected to run the simulation at 259.5mph. Consequently, the simulation of the model with velocity at 259.5mph we receive promising results. The residual follows a rather fluctuating path in the beginning but follows more of a linear pattern during the end. The vector and scalar representation depicts the flow of air in terms of streamlines and vectors respectively. The former represents the pressure variation from low to high at the edges of the wing. This is also same for the velocity streamlines where it decreases when it hits the wing and flow towards the exit. Drag and Downforce follows a linear trend post 50 iterations. This is due to the turbulent parameters set before the simulation. Overall, this study has been effective in learning about mesh independency, residuals, reports and the knowledge required to conduct such a study.